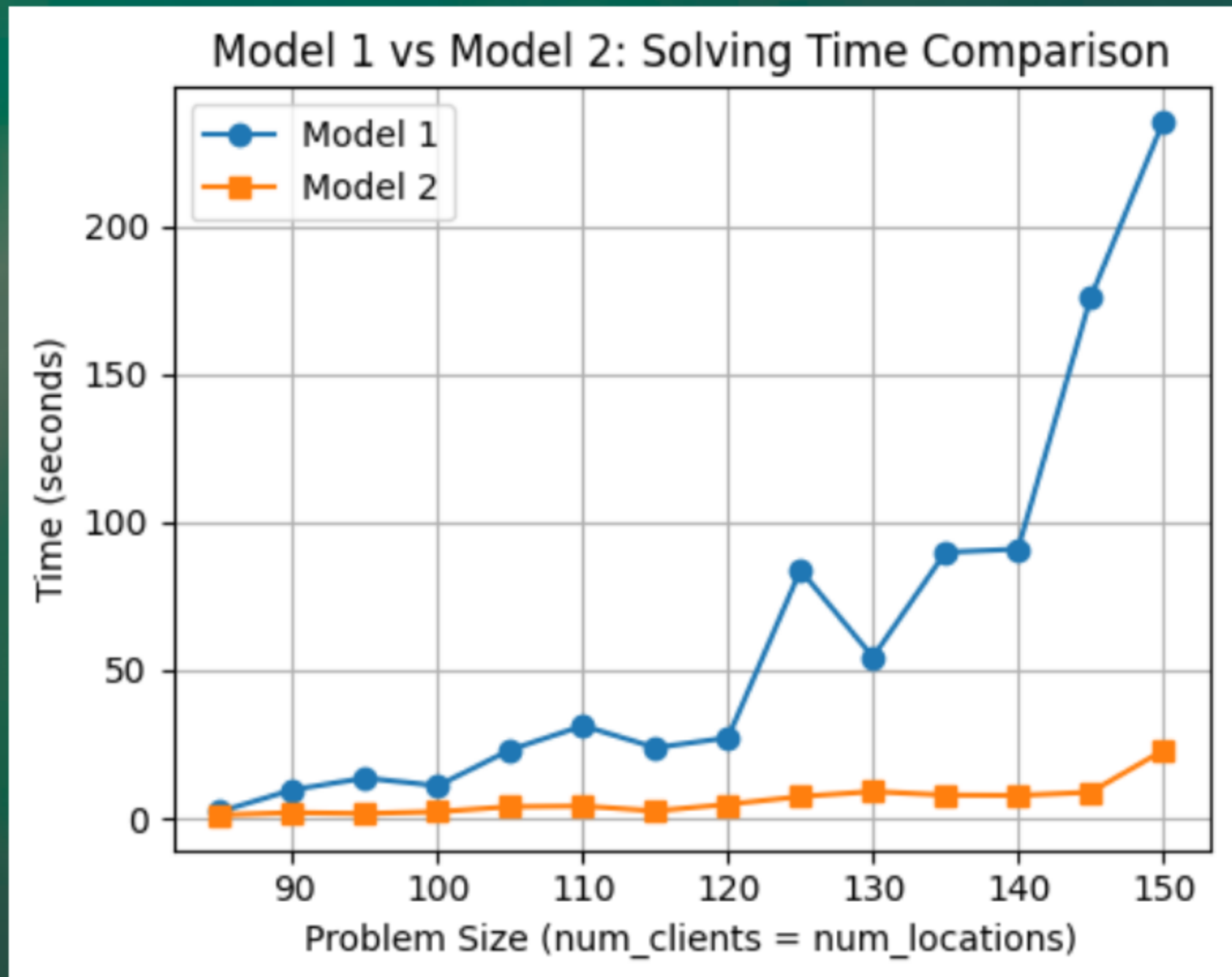


Know the trick = 10x better performance



The Art of Mathematical Modeling

next page →



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SOLUTIONS

Example

Maximum Coverage Problem

Objective:

-  Place facilities
-  Cover as many clients as possible



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Constraints:

-  Every facility serves different clients
-  You can only place B facilities in total



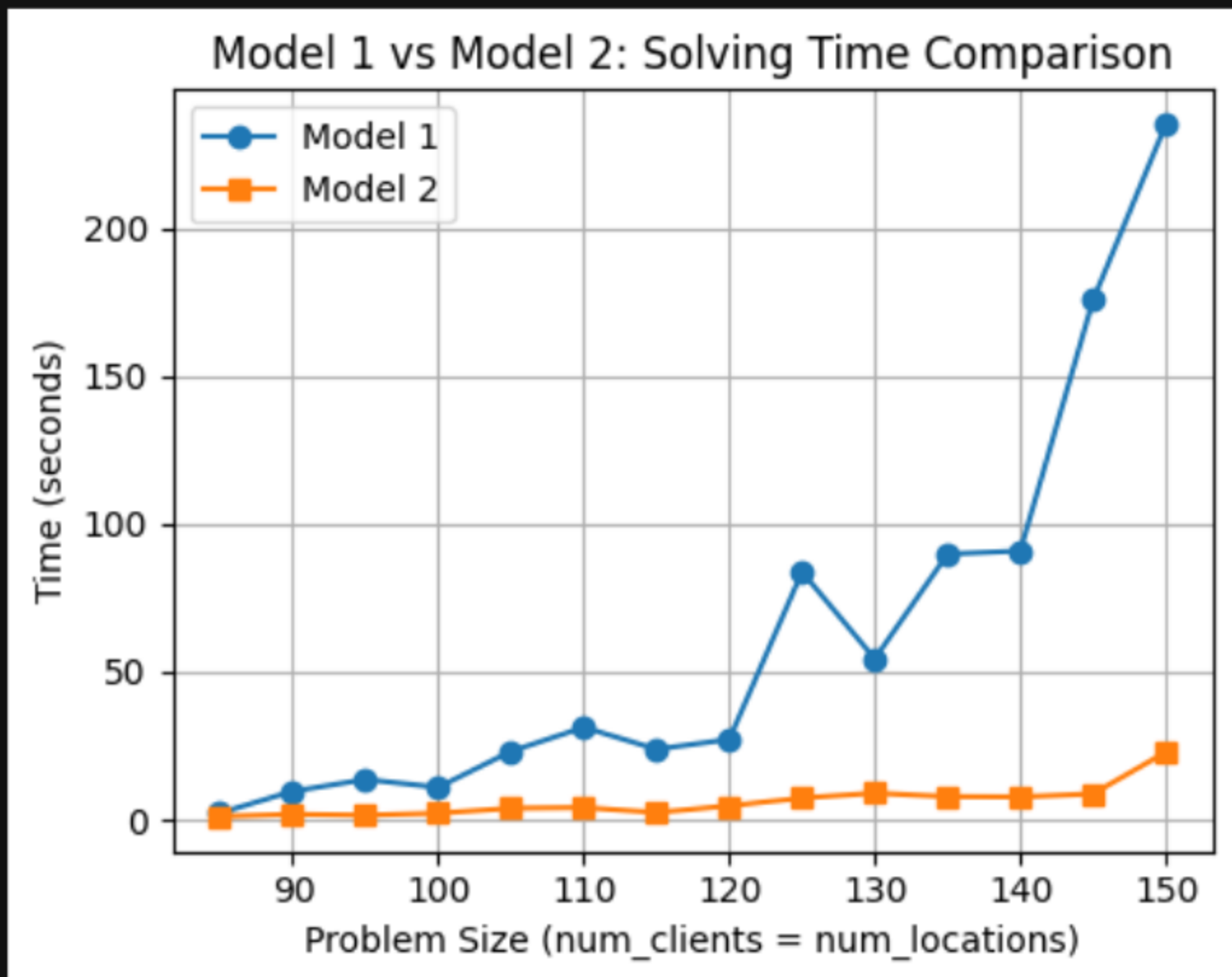
The catch

-  You can formulate this problem in two different ways.
-  Either, when solved, yields the optimal placement of facilities.
-  But one formulation is solved much faster



Model comparison (1/2)

Model 2 runs more than 10x faster and scales much better.



Model comparison (2/2)

	Model 1	Model 2
Variables	$O(n \cdot m)$	$O(n + m)$
Constraints	$O(n \cdot m)$	$O(m)$

where $n = |N|$ (number of facilities) and m (number of clients).



Live Session and more Modeling Know-How:

Optimization4All

HANDS-ON MODELLING SERIES
March 11TH 5PM - 7PM CET on Zoom



JENS-PETER JOOST
OR Engineer

Hands-On MIP Modeling: From Simple Formulations to Scalable Solutions

Prerequisites:

- Basic Python programming familiarity.
- Basic OR knowledge about variables, constraints and objective function.

Collaborators: Prajit Adhikari, Siva Visvesvaran and Ha Nguyen



Registration Link in the comments!



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SOLUTIONS

Model 1

Decision Variables:

$$x_i = \begin{cases} 1 & \text{if facility is placed at location } i \in N \\ 0 & \text{otherwise} \end{cases}$$

$$y_{ij} = \begin{cases} 1 & \text{if facility } i \text{ serves client } j \\ 0 & \text{otherwise} \end{cases} \quad i \in N, j \in S_i$$

Objective:

$$\max \sum_{i \in N} \sum_{j \in S_i} y_{ij}$$

Subject to:

$$\sum_{i \in N} x_i \leq B$$

$$\sum_{i: j \in S_i} y_{ij} \leq 1 \quad \forall j$$

$$y_{ij} \leq x_i \quad \forall i \in N, j \in S_i$$

where S_i is the set of clients that facility i can serve.



Model 2

Decision Variables:

$$x_i = \begin{cases} 1 & \text{if facility is placed at location } i \in N \\ 0 & \text{otherwise} \end{cases}$$

$$z_j = \begin{cases} 1 & \text{if client } j \text{ is served} \\ 0 & \text{otherwise} \end{cases}$$

Objective:

$$\max \sum_j z_j$$

Subject to:

$$\sum_{i \in N} x_i \leq B$$

$$z_j \leq \sum_{i \in S_j} x_i \quad \forall j$$

where S_j is the set of facilities that can serve client j .